

CHE 435/525 *Process Systems Analysis and Control*

Lecture 3: **Reactor Models, Dynamic Simulation,**
and Their Nonlinear Behaviors

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Discuss & Recapture

- How does a process model usually look like?

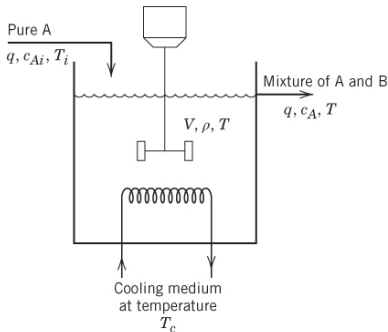


- What assumptions do we typically make?
- How to write out the model quickly and accurately?

Learning objectives of this lecture

- ① Model tanks (CSTRs) involving **chemical reactions**
 - ▶ Challenge: Deal with thermodynamics of a mixture
- ② **Simulate** the reactor model with Matlab
- ③ See the complexity in the behaviors of a **nonlinear system**

Example 5: CSTR (non-isothermal)



- Assumptions

- 1 Constant volume holdup ($q_{\text{in}} = q_{\text{out}} = q$)
- 2 Constant ρ and C_P (independent of c_A): A and B have equal ρ and C_P and form an ideal mixture

Mass balance

- Molar balance for A:

$$\frac{d(Vc_A)}{dt} = qc_{Ai} - qc_A - rV$$

- Reaction kinetics:

- ▶ Rate law: $r = kc_A$
- ▶ Arrhenius law: $k = k_0e^{-E/RT}$ (k_0, E, R : constants)

⇒

$$\frac{dc_A}{dt} = \frac{q}{V}(c_{Ai} - c_A) - k_0e^{-E/RT}c_A$$

- Why don't we need to do a mass balance for B?
- What if this is a multi-reaction system?

Energy balance

- Assuming $U = H$ for liquid:

$$\frac{dH}{dt} = \dot{H}_{\text{in}} - \dot{H}_{\text{out}} + Q$$

- H : total enthalpy, $\dot{H}_{\text{in}}$, $\dot{H}_{\text{out}}$: enthalpy flow rate
- Heat exchange: $Q = -UA(T - T_c)$ (U, A : constants)

- Enthalpy-temperature relation: **Not so easy!**

- We can't simply use $\hat{H} = C_P T$. **The molar enthalpy of A and B can't be both zero at the reference temperature.**
- We may set $\hat{H}_A = C_P T$. Then $\hat{H}_B = C_P T + \Delta\hat{H}_r^\circ$ (standard molar enthalpy change of reaction)

$$H = V c_A \hat{H}_A + V c_B \hat{H}_B = V [c_A C_P T + c_B (\Delta\hat{H}_r^\circ + C_P T)]$$

- Then,

$$\frac{dH}{dt} = V \left[\frac{dc_A}{dt} C_P T + \frac{dc_B}{dt} (\Delta\hat{H}_r^\circ + C_P T) + (c_A + c_B) C_P \frac{dT}{dt} \right]$$

Energy balance (cont.)

- Note: $c_A + c_B = c_{Ai}$, $dc_A = -dc_B$.

$$V \left[\frac{dc_A}{dt} (-\Delta \hat{H}_r^\circ) + c_{Ai} C_P \frac{dT}{dt} \right] = \dot{H}_{in} - \dot{H}_{out} - UA(T - T_c)$$

- How about $\dot{H}_{in, out}$?

$$\begin{aligned} \dot{H}_{in} &= q c_{Ai} C_P T_i, \quad \dot{H}_{out} = q [c_A C_P T + c_B (\Delta \hat{H}_r^\circ + C_P T)] \\ \dot{H}_{in} - \dot{H}_{out} &= q [c_{Ai} C_P (T_i - T) - (c_{Ai} - c_A) \Delta \hat{H}_r^\circ] \end{aligned}$$

- Plug in mass balance: $\frac{dc_A}{dt} = \frac{q}{V} (c_{Ai} - c_A) - k_0 e^{-E/RT} c_A \Rightarrow$

$$V c_{Ai} C_P \frac{dT}{dt} = q c_{Ai} C_P (T_i - T) + V k_0 e^{-E/RT} c_A (-\Delta \hat{H}_r^\circ) - UA(T - T_c)$$

Energy balance ... Finally!

$$\frac{dc_A}{dt} = \frac{q}{V}(c_{Ai} - c_A) - k_0 e^{-E/RT} c_A$$

$$\frac{dT}{dt} = \frac{q}{V}(T_i - T) + \frac{k_0 e^{-E/RT} c_A (-\Delta \hat{H}_r^\circ)}{C_{PCAi}} - \frac{UA(T - T_c)}{V C_{Ai} C_P}$$

- How to make sense of this model?

- ▶ V/q : residence time, $(q/V)(\text{in} - \text{out})$: effect of inlet
- ▶ $k_0 e^{-E/RT} c_A$: reaction rate [=] mol/m³/s
- ▶ $(-\Delta \hat{H}_r^\circ)$: heat released from reaction [=] J/mol
- ▶ $UA(T - T_c)$: cooling rate [=] J/s
- ▶ VC_{PCAi} : total heat capacity [=] J/K

Example 6: Isothermal CSTR with multiple reactions)

- Setting (Gray-Scott model):

① $q_i = q = \text{const.}$, $V = \text{const.}$

② Reactions: $A + 2B \xrightarrow{k_1} 3B$, $B \xrightarrow{k_2} C$, both are elementary reactions

- Mass balances:

$$V \frac{dc_A}{dt} = qc_{Ai} - qc_A - \text{I?}$$

$$V \frac{dc_B}{dt} = qc_{Bi} - qc_B + \text{II?}$$

- $\text{I} = Vk_1c_Ac_B^2$, $\text{II} = V((3 - 2)k_1c_Ac_B^2 - k_2c_B)$

Summary on process modeling

- ① Start with mass and energy balances
 - ▶ Based on the description, include all relevant terms [CHE 205/225]
 - ▶ Transport phenomena are handled in a simplified way here. (In a real setting, use PDEs or more accurate correlations.) [CHE 311/312]
- ② Relate variables to T , P , compositions, and mass/moles
 - ▶ Thermo: enthalpy, chemical potential (fugacity), etc. [CHE 315/316]
 - ▶ Reaction: rate law [CHE 446]
- ③ Simplify... Just be patient!
 - ▶ Interpret the model, verify the units, check DoF

Simulating the CSTR with Matlab

- Consider the isothermal reactor

$$\frac{dc_A}{dt} = (q/V)(c_{Ai} - c_A) - k_1 c_A c_B^2$$
$$\frac{dc_B}{dt} = (q/V)(c_{Bi} - c_B) + k_1 c_A c_B^2 - k_2 c_B$$

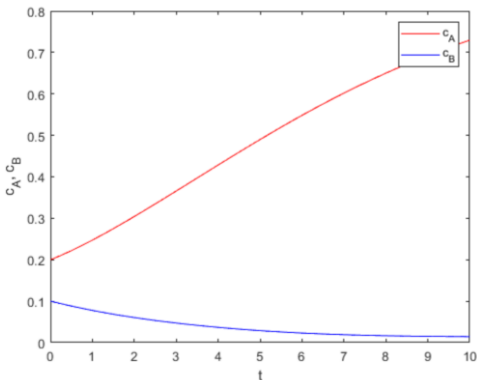
where $q/V = 0.15 \text{ min}^{-1}$, $c_{Ai} = 1 \text{ mol/L}$, $c_{Bi} = 0.0667 \text{ mol/L}$,
 $k_1 = 40 \text{ L}^2 \cdot \text{mol}^{-2} \cdot \text{min}^{-1}$, $k_2 = 1 \text{ min}^{-1}$

- Chemical processes are almost always **nonlinear systems!**
 - ▶ No analytical solution should be expected
 - ▶ The behavior may be complicated ... let's see how

Codes: Mind the syntax

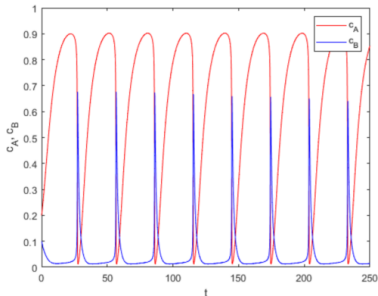
- Suppose $c_A(0) = 0.2$, $c_B(0) = 0.1$

```
sim = @(t, x) [0.15*(1-x(1)) - 40*x(1)*x(2)^2;  
              0.15*(0.0667-x(2)) + 40*x(1)*x(2)^2 - 1*x(2)];  
tspan = 0:0.05:10;  
x0 = [0.2; 0.1];  
[t, x] = ode45(sim, tspan, x0);  
plot(t, x(:,1), 'r', t, x(:,2), 'b');  
xlabel('t'), ylabel('c_A, c_B'), legend({'c_A', 'c_B'});
```



** Limit cycle behavior

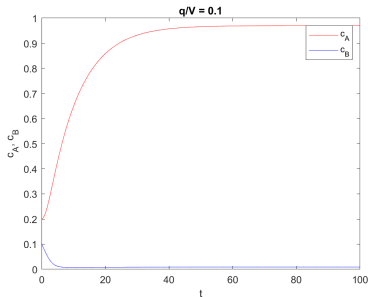
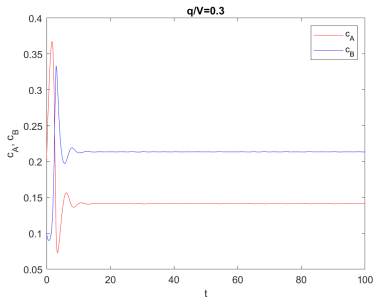
- Simulate for 250 minutes... The system is attracted to (non-sinusoidal) periodic oscillations – called a **limit cycle**



- We did not force the system to oscillate, but only replenish reactants
- Think about **biological clocks**, **cell mitosis**, etc.

** Bifurcation

- Increase q/V to 0.3 (a smaller tank) or decrease to 0.1 (a larger tank)



- The reactor can exhibit **high-conversion steady-state**, **cyclic**, **low-conversion steady-state** behaviors, **sensitive** to parameters (operating conditions) (in this case: residence time)

Nonlinear Systems v.s. Linear Systems

- Generally speaking, **nonlinear ODEs** can have highly complex traits
 - ▶ An intriguing area of scientific research
- This course shall focus on linear systems (from next lecture)
 - ▶ Reason: Linear systems are easier to study, while nonlinear systems require much more mathematics
 - ▶ Justification for this laziness: Nonlinear systems are approximately linear, if operated near a **steady state**
 - ▶ Tools: Laplace transform and transfer function